Highlights

The Boussinesq–compressible gap in steady two-dimensional buoyancy-driven enclosure convection is linear to leading order in the relative temperature contrast

- Boussinesq–compressible gap scales as $\varepsilon^{1.02}$ in velocity over 627 paired solves
- Temperature contrast alone breaks the symmetry; a 24$\times$ Ma change is below resolution
- Correction factorises as $D = \varepsilon P(X)$: networks learn only the spatial shape
- One inference takes 6.3 ms; per-pair cost falls 2.9$\times$ and 13$\times$ for a 20$\times$5 scan
- At $\varepsilon = 1.2$ integral Nu shifts 0.54%; a learned prior cuts field error 70$\times$ but wall-Nu error stays 2.6%